

NIMA HEMATI

✉ nima.hemati@mail.mcgill.ca

☎ 514-699-1401

in LinkedIn

🐙 GitHub

PROFILE

Mechanical engineer (B.Eng. + M.Sc. Surgical Robotics, software engineering minor) currently leading the technical side of a soft-robotic surgical product at Taurus and founding technical lead on a stealth healthcare startup. Have spent the last few years building robotics platforms across hardware, software, and AI, and a fair amount of time pitching them to investors, clinical partners, and trade-show audiences. Looking for product or technical leadership roles in deep tech.

SUMMARY OF SKILLS

- Comfortable taking ambiguous executive or clinical input and turning it into a concrete spec engineers can build against; used to defining KPIs and weighing trade-offs across performance, reliability, cost, and schedule.
- Have spent considerable time pitching and demoing deep-tech products to **investors, clinical partners, and trade-show audiences**.
- Hands-on across mechanical CAD/CAE (**Dassault Systèmes, Ansys, Siemens**), real-time software (**C++, C#, Qt**), and AI/ML (**PyTorch, TensorFlow**), including in regulated industries (**ISO 13485, IEC 62304, ISO 14971**).
- Fluent in English and French.

PROFESSIONAL EXPERIENCE

FOUNDING TECHNICAL LEAD

📍 Montréal, QC

STEALTH HEALTHCARE STARTUP

📅 Mar. 2026 – Present

- Lead all technical planning and development for a healthcare asset-management platform; primary technical voice when pitching to investors alongside the founding team.
- Architecting a full-stack web application (**ViteJS, React, Spring Boot, SQL**) plus the **IoT and networking** layer for asset tracking.

ROBOTICS DEVELOPER

📍 Westmount, QC

TAURUS SURGICAL INC.

📅 Aug. 2024 – Present

- Lead the technical side of a **soft-robotic catheter** program. Make system-level hardware/software architecture calls from executive requirements, then build out the mechanical (**Fusion 360, SolidWorks, SLA/FDM**) and software pieces.
- Primary technical demonstrator for **investors, potential collaborators, and trade-show audiences**; spend considerable time positioning the product against alternatives and walking stakeholders through the system.
- Run **Agile** delivery (**Git, JIRA**) across all of the company's software, including the **C++/QT/C#** digital-twin and real-time control stack that drives a soft and rigid robot in tandem.
- Handle the regulatory side (**ISO 13485, IEC 62304, ISO 14971**): risk analysis, CAPA, and technical documentation for Canadian and US MedTech.

RESEARCH ASSISTANT

📍 Montréal, QC

RI-MUHC - SUPER LAB

📅 May. 2023 – Present

- Led three different surgical-robotics projects from concept through working demos: laparoscopic multi-robot teleoperation (**Kinova Gen 3**), semi-autonomous orthopaedic resection (**KUKA LBR Med**), and a tendon-driven soft-robotic catheter (current MSc thesis). Regularly demoed each to **investors, clinical collaborators, and lab visitors**.
- Built the real-time control stacks (**C++, C#**), Unity-based 3D planning UIs, and an AI-based (**VAE + Transformer**) kinematics model that runs real-time control on the catheter.
- Two first-author publications (poster + podium) at the **Hamlyn Symposium on Medical Robotics**; 2× Best Podium Presentation winner (McGill Surgical & Interventional Sciences and Fraser Gurd Research Days).

FOUNDER

MYRADIO

📍 Montréal, QC

📅 Sep. 2025 – May 2026

- Founded **MyRadio**, a patient-specific 3D-printing service for orthopaedic and cardiovascular surgery at MUHC. Ran the pilot end-to-end as a solo founder: established direct clinician relationships, segmented CT/MRI imaging in **3D Slicer**, modified geometry in **Fusion/SolidWorks**, and printed via **SLA/FDM**.

ROBOTICS SOFTWARE CONTRACTOR

HOLORAY

📍 Montréal, QC

📅 Nov. 2025 – Jan. 2026

- Lead demonstrator for HoloRay's live cross-continental teleoperation demo with **Haply Robotics** at **CES 2026**; presented on the show floor, connected with prospective customers and partners, and tracked inbound leads.
- Built the **C++/websocket** robot-control and path-planning software for the demo: a Haply Inverse3 haptic device in Las Vegas drove a robot in Montréal in real time, networked over HoloRay's FDA-approved **HoloPod IoT** device.

TECHNICAL DIRECTOR/TRANSMISSION LEAD

MCGILL BAJA RACING

📍 Montréal, QC

📅 Sep. 2019 – May. 2023

- Set the team's multi-year technical direction by translating prior-season competition results (acceleration, suspension and traction, endurance) into specific subsystem requirements.
- Built a multi-stage simulation suite (**Excel, MATLAB, Python**) running from event KPIs to vehicle dynamics to CVT geometry; predicted a **65 kph** top speed against **68 kph** measured on track and fundamentally changed how the team designs powertrains. Still in use today.
- Team finished **8th out of 100 international teams** (best in a decade) in the first season running the new powertrain workflow; spearheaded weekly subsystem meetings and presented to judges at design competitions.
- Primary liaison with local manufacturers; machined aluminum/steel parts on **CNC mill/lathe (MasterCAM)** and CNC-routed moulds for **carbon-fibre body panels** (Body Panels Lead, **Siemens NX**, before Tech Director role).

SYSTEMS ENGINEERING INTERN

MAYA HTT

📍 Montréal, QC

📅 May. 2022 – Aug. 2022

- Worked directly with enterprise **aerospace** clients to scope and deliver **CAE solutions: random and sinusoidal vibration analyses (Simcenter FEMAP, Maya SATK)** and **system-level powertrain/thermal models (Simcenter Amesim)**; also pitched the **Simcenter** product portfolio to clients.

QUALITY ASSURANCE INTERN

MARTINREA INTERNATIONAL INC.

📍 Tillsonburg, ON

📅 May. 2021 – Aug. 2021

- Built an **IoT** system for AI-based part QA: trained **ML models (Python, TensorFlow, OpenCV)** deployed on **Raspberry Pi** edge devices, with a **Flask/HTML/JS** web client backed by **SQL**; reconciled **\$48M** of missing tooling tags via Excel.

EDUCATION

MCGILL UNIVERSITY

MASTER OF SCIENCE, SURGICAL ROBOTICS

📍 Montréal, QC

📅 Sep. 2024 – May. 2026

- GPA: 4.00/4.00. Awards: Graduate Excellence award, FRQNT Scholar.
- Thesis: Robust, Optimal Control of the Dynamics of a Tendon-Driven Robotic Catheter

MCGILL UNIVERSITY

BACHELOR OF ENGINEERING, HONOURS MECHANICAL ENGINEERING

📍 Montréal, QC

📅 Sep. 2019 – May. 2024

- GPA: 3.73/4.00; Minor in Software Engineering. Awards: Class of 1983 scholarship, Dean's Honor List, 3× TEAM Mentorship award.
- Thesis: Trajectory Planning and Control for a Hydraulic Log-Loading Manipulator
- Activities: Founder of **McGill aCADemy** CAD tutoring service (2020-2022); McGill Baja Racing (2019-2023).